C2 # 2, 3, 5, 9, 10, 11, 12, 13, 15, 16

<u>Thm</u>: (Inclusion – Exclusion)

$\mathcal{P}$: $P_1, P_2, \dots P_m$ $\quad S: x_1, x_2, \dots x_n$

$i \in [m]$ $\quad A_i = \{x \in S \mid x \text{ has property } P_i\}$

$T \subseteq [m]$, $A_T = \bigcap_{i \in T} A_i$

$$\left|\bigcap_{i \in [m]} \overline{A_i}\right| = \sum_{T \subseteq [m]} (-1)^{|T|} |A_T|$$

$$|A_T| = |S| - \sum_{i=1}^{n} |A_i| + \dots$$

<u>Pf</u>: $\forall x \in S$, $J_x = \{p \in \mathcal{P} \mid x \text{ has property } P\}$

$\# |J_x| = \phi \vee \# |J_x| \neq \phi$

W.L.O.G, $J_x = \{P_1, P_2, \dots P_k\}$ $k \geq 1$

x contributes to LHS 0 times

LHS: $1 - k + \binom{k}{2} - \binom{k}{3} + \dots + (-1)^k \binom{k}{k} = (1-1)^k = 0$ times

<u>Eg</u> Derangement $D_n = |\{\delta \in S_n \mid \delta_i \neq i, \forall 1 \leq i \leq n\}|$ Derive D_n

<u>Sol</u>: $T = \{i_1, i_2, \dots i_k\}$ $\delta_{i_1} = i_1, \dots \delta_{i_k} = i_k$

$|A_T| = (n-k)!$

$$D_n = \sum_{T \subseteq [n]} (-1)^{|T|} |A_T| = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$$

<u>Eg</u> # of surjections from $[n]$ to $[m]$.

<u>Sol</u>: S: all maps from $[n]$ to $[m]$.

P_i: $i \in [m]$ is not mapped

$\forall 1 \leq i \leq m$, $|A_i| = (m-1)^n$, $|A_i \cap A_j| = (m-2)^n$

$$\# = \sum_{T \subseteq [m]} (-1)^T |A_T| = \sum_{k=0}^{m} (-1)^k \binom{m}{k} (m-k)^n = \sum_{i=0}^{m} (-1)^{m-i} \binom{m}{i} i^n$$

$$m!\, S(n,m)$$

Eg Fix k, n $\quad |\{\delta \in S_n \mid \delta \text{ has exactly } k \text{ fixed pts}\}| = ?$